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Incentive Schemes for Centralized Material Planning and Allocation With Asymmetric Information in Construction Supply Chain

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ABSTRACT In construction projects, a centralized management mode can effectively address the material management issues. Specifically, the Material Management Department (MMD) is responsible for material purchase, storage, and allocation based on the demand information reported by contractors. However, material demand are best observed by the contractors. They may manipulate the material takeoff (i.e. misrepresent the material demand) to increase the expected material allocation and avoid the project delay. Motivated by this situation, this paper focuses on the incentives to elicit truthful reporting from contractors. Cases of having prior information and no prior information about the material takeoff, as well as priority allocation and proportional allocation policies are considered. Under the case of having prior information, an incentive scheme is designed based on the Groves payment rule, while under the case of no prior information, two incentive schemes are designed based on the proper scoring rules. The comparison and numerical analysis show that the total three incentive schemes can induce the MMD to choose a system-optimal inventory plan, as well as satisfy the participation constraints of contractors. Besides, the results also demonstrate that prior information is only the prerequisite for implementing the Groves incentive scheme and cannot improve the performance of the whole system in other aspects.

INDEX TERMS Construction supply chain, material management, information asymmetry, incentive scheme.

I. INTRODUCTION

Statistically, the cost of materials can constitute 50% to 60% of the total cost of construction projects [1], [2], [3], and efficient material management can help improve the schedule, cost, and quality performance. In some mega projects, such as the Three Gorges Project in China, a centralized management mode is implemented to address the issues surrounding some bulk materials required by multiple activities with massive demand [4], [5]. Specifically, under this mode, the MMD receives demand forecasts from contractors, and then makes plans for centralized material inventory and allocation. This mode has been demonstrated as an effective mitigation

strategy to cope with the impact of disruptive events [6], [7], [8].

However, all the material planning issues mentioned above rely on the contractors' material takeoff. In construction projects, material takeoff is affected by the project design, construction techniques, workers' skill levels, etc. Contractors usually are more professional and have superior information about these factors [9]. To increase the expected material allocation and decrease the corresponding project delay cost, contractors may overstate the material takeoff. The distorted material takeoff could lead various deficiencies in materials management and deteriorate the performance [10], [11]. This issue is caused by the asymmetry material demand information between the MMD and contractors. Solving the problem of asymmetric information and ensuring the accuracy of

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material takeoff is vital to the performance of construction supply chain [10], [12].

Information asymmetry has been a crucial issue in project management for a long time [13], [14]. Some participants are usually more informed about some facts, while others are less informed, just like the material demand information in our context [15]. In engineering practice, it mainly relies on the supervision mechanism and Building Information Modeling (BIM) [16], [17] to solve this problem. For instance, with the help of BIM quantity take-off services by reviewing drawings and specifications, the MMD can get a rough idea of the amount of materials required to complete the project. However, the information is not necessarily accurate. It can only alleviate information asymmetry to a certain extent, but cannot eliminate it completely. The contractors still can overstate the material takeoff and induce the MMD to purchase more material than they need. Motivated by this, proper incentives need to be well-designed to completely eliminate the information asymmetry between MMD and contractors.

Information asymmetry issues have been extensively examined in the literature both on construction project management and manufacturing/retail supply chain management. Xiang et al. [18] summarized 30 phenomena of asymmetric information in construction projects, such as the contractor's technical and management capabilities, understanding of construction intent and construction conditions, and construction progress control capability. In terms of these information asymmetry issues, incentives are considered effective solutions. Incentives can stimulate consistency of information and align interests among project participants. For example, the time-based and cost-sharing incentive contracts are designed to deal with the moral hazard caused by contractor's unobservable work rate [19], [20]. It is a pity that industry professionals and researchers are aware of the asymmetric material demand information, but do not often openly discuss about the corresponding incentives. From a modeling perspective, the incentive design problem in the context of construction projects is related to some classic problems studied in the literature on manufacturing/retail supply chain management. The well-known buyback contract [21], wholesale price contract [22], [23], [24], quantity discount contract [25], [26], revenue sharing contract [27], [28], and two-part tariff contract [29] are all elaborately designed to induce asymmetric demand information between buyers and sellers. However, the MMD and contractors in construction projects are not of buying and selling relationships. These incentives and contracts cannot directly address the material demand information asymmetry.

The main contribution lies in the fact that we are dealing with a different and new problem. The aforementioned works mainly study the incentive schemes in the field of production-distribution-retail, which, though effective, cannot be directly used in the construction domain. The purpose of this work is to suggest incentive schemes for solving the asymmetric material demand information in construction supply chain. In this work, a simple and highly stylized single-period model

that includes one MMD and n contractors is examined. The contractors report their material takeoff, while the MMD purchases material from an external supplier and allocates it to the contractors as per the announced policies, such as the priority policy and proportional policy. Meanwhile, both the cases of having prior information and no prior information about the material demand are also considered. On this basis, we aim to design incentive schemes to solve the material demand information asymmetry issues with different cases of prior information and allocation policies consideration. Specifically, several incentive schemes are going to be designed based on the Groves payment rule and proper scoring rule. The optimal decisions of MMD and contracts will be examined for investigating how the designed incentive schemes solve the information asymmetry and coordinating issues.

The rest of the paper is organized as follows. Section II presents the literature review, and Section III specifies the methodologies for this study. The detailed model symbols, assumptions and formulation are outlined in Section IV. Section V presents the three incentive schemes, and Section VI conducts comparison analysis and numerical study of the incentive schemes. The last section summarizes our research findings and future directions.

II. LITERATURE REVIEW

In construction management literature, some studies focus on the process of modeling and on the optimization of the construction material supply chain. Walsh et al. [30] strategically positioned material inventory to balance the contractor's desire to have materials on site early and the supply chain benefits from inventory reduction. Tserng et al. [31] introduced the Vendor Managed Inventory (VMI) philosophy into construction projects and developed a model minimizing the integrated inventory cost of the supply chain. Georgy and Basily [32] developed a computerized tool for optimizing the delivery and inventory of materials, as part of a comprehensive material management system in construction projects. Xu et al. [33] considered the recurrent projects with random material delays, and developed an integrated approach to jointly optimize material safety stock and project crashing decisions. Lu et al. [6] focused on the inventory control problem of construction material and proposed general safety stock determination approaches under nonstationary stochastic demand and random supply yield. All the above models can support informed decisions and bring competitive advantages. But they alone cannot ensure project success because they ignore the participants' self-interest. Incentives are needed to balance the participants' potential interest conflicts and guarantee performance.

Incentives play an important role in ensuring the time, cost, and quantity performance of construction projects. According to Ashley and Workman [34], incentives can motivate a contractor to adopt the client's project objectives. Bower et al. [35] pointed out that correctly applying incentives can motivate both client and contractors to focus on the

appropriate business objectives and lead to successful project results. Meng and Gallagher [36] provided empirical evidence for the significant impact of incentive mechanisms on project performance. Besides, as stated by Xiang et al. [18] and Pesek et al. [37], information asymmetry is a widespread problem in construction projects. From the perspective of information economics, asymmetric information leads to opportunistic behavior, which is the primary cause of loss and risk in construction projects [39]. Under this circumstance, incentives are even more important.

At present, numerous studies focus on incentive design under asymmetric information setting in construction projects. Ceric [12] and Schieg [40] pointed out that incentives (bonuses) is an effective strategy. Bayiz and Corbett [19] considered a construction project in which the manager cannot observe the subcontractors work rate, and derived the optimal linear incentive contracts to induce the subcontractors to exert appropriate effort. Kwon et al. [20] also studied project-contracting issues under unobservable contractor's work rate and investigated fixed price, cost-based and time-based contracts, respectively. It was shown that, with carefully chosen parameters, time-based and cost-sharing contracts can achieve optimal channel coordination. Shi et al. [41] analyzed the double moral hazard caused by asymmetric information in construction projects, and designed an optimal risk-sharing scheme. It demonstrated that the risk-sharing scheme plays an important role in coordinating both parties to choose efficient strategies. Yao et al. [42] considered the asymmetric cost information of contractor, and designed a menu of incentive contracts to coordinate the profits of a risk-averse owner and a risk-neutral contractor. However, these studies mainly focus on the asymmetric information of project schedule, cost, and quality. The incentives that cope with the asymmetric information on material demand is still missing.

In manufacturing/retail supply chain management, incentives for asymmetric demand information has been extensively studied in the literature. Numerous well-known contracts are adopted to solve the information asymmetry. Burnetas et al. [43] studied single-period supply contract issues under asymmetric demand information and investigated quantity discount contracts for a supplier to elicit private demand information from a newsvending buyer. Gan et al. [44] considered a drop-shipping supply chain in which the retailer receives a customer's order and the supplier keeps inventory and fulfills orders, and proposed commitment-penalty contracts to induce the retailer with superior information to report its truthful demand information. Babich et al. [21] considered the coordination problem between a powerful supplier and a more informed retailer, and derived an optimal buyback contract and showed the condition under which the supplier extracts all the surplus. Heese and Kemahlioglu-Ziya [27] considered a single-supplier single-retailer supply chain, and designed a revenue sharing contract to prevent the retail from distorting the private demand information. Chu et al. [22] and Lee and

Liang [24] studied a two-echelon supply chain comprised of a retailer who sources a product from a manufacturer and designed wholesale price contract to induce the market demand truthfully. Besides, some other less common contracts are also adopted to address coordination issue of production-distribution under asymmetric demand information setting, such as advance sale [45] and franchise fee [46].

Roughly speaking, the buyback, wholesale price, quantity discount, revenue sharing, etc., are effectively used to address the asymmetric demand information in manufacturing/retail supply chains. Nevertheless, the relationship between MMD and contractors regarding the material supply is not buying and selling. The MMD purchases material from external suppliers and provides to the contractors for use. The above contracts cannot directly be used to address the information asymmetry in this context. To fill this research gap, this study will investigate the incentives considering asymmetric material demand information in construction supply chain.

III. METHODOLOGIES

A. GROVES MECHANISM

Groves mechanism is one of the main achievements in this field of incentive design theory. Its core idea is to tie each participant's private profit with the total profit of the whole system. Therefore, each participant's decisions coincide with the optimal decisions of the whole system. A briefly introduction about this mechanism is presented here.

There are n agents, and each one has a set of possible types Θ_i . Let Y be the set of all possible alternative outcomes, and s be the selection rule $\Theta \rightarrow Y$. Each type $\theta_i \in \Theta_i$ of any agent i has a valuation function $f_i(s(\theta'_i, \theta'_{-i}), \theta_i)$. The selection rule s is outcome efficient if it always realizes the maximum gains from trade in θ , as [47], [48]

$$s = \max_s \sum_{i=1}^n f_i(s(\theta), \theta_i) \quad (1)$$

Define a mechanism $\langle s, t \rangle$ with a trading charge e_i imposed on i as

$$e_i(\theta) = F(\theta) - f_i(s(\theta), \theta_i) - t_i(\theta) \quad (2)$$

where $F(\theta)$ represents the whole gains from trade, i.e. the sum of $f_i(s(\theta), \theta_i)$. Under this mechanism, each agent i receives a total payoff equal to the whole gains from trade, $F(\theta)$, minus a trading charge $e_i(\theta)$.

According to Makowski and Mezzetti [47], $\langle s, t \rangle$ is a Groves mechanism if (i) s is outcome efficient, and (ii) the trading charges are lumps sums, i.e. $\forall i, \forall \theta_{-i}, \forall \theta'_i$, and $\forall \theta''_i$,

$$e_i(\theta_{-i}, \theta'_i) = e_i(\theta_{-i}, \theta''_i) \quad (3)$$

$\langle s, t \rangle$ is a Groves mechanism in expectations if (i) s is outcome efficient, and (ii) the trading charges are lumps sum in expectations, i.e. $\forall i, \forall \theta'_i$, and $\forall \theta''_i$,

$$E_{-i}[e_i(\theta_{-i}, \theta'_i)] = E_{-i}[e_i(\theta_{-i}, \theta''_i)] \quad (4)$$

Moreover, Makowski and Mezzetti [47] and Narahari et al. [48] pointed out the well-known equivalences

between Groves mechanism (respectively, Groves mechanism in expectations) and dominant strategy (respectively, Bayesian) mechanism. Specifically, the mechanism $\langle s, t \rangle$ is dominant strategy incentive compatible and outcome efficient if and only if $\langle s, t \rangle$ is a Groves mechanism; The mechanism $\langle s, t \rangle$ is Bayesian incentive compatible and outcome efficient if and only if $\langle s, t \rangle$ is a Groves mechanism in expectations. These strong properties bring the popularity of Groves mechanism.

As stated by D'Aspremont and Gerard-Varet [49], a dominant strategy equilibrium is consistent with “complete ignorance”. That is, a dominant strategy equilibrium is a Bayesian equilibrium under any prior information. By contrast, the Bayesian equilibrium relies on the prior information. So the first thing that comes to mind is to design a dominant strategy incentive compatible mechanism to solve the information asymmetry issue. However, according to the finding by Arrow [50], Holmstrom [51] and Walker [52], budget balance (which is one of the desirable requirements of incentive schemes design in this study) and outcome efficient dominant strategy mechanisms rarely exist, while budget balance and outcome efficient Bayesian mechanisms always exist. Hence, we aim to design a Groves incentive scheme in expectations under the scenario with prior information (see part A in Section V).

B. SCORING-RULE-BASED MECHANISM

Scoring rules are widely used to assess and reward a prediction given in probabilistic form [53]. Given the reported probability distribution and the actual event, a score can be calculated. This score represents the accuracy of the reported probability distribution. Specifically, a scoring rule is called strictly proper if the score can be minimized (in expectation) by reporting the true information. That is, when the actual event ε is drawn from the probability distribution Ω (which can be denoted as $\varepsilon \sim \Omega$), the expected score is minimized if it reports Ω rather than any other distribution Ω' :

$$E_{\varepsilon \sim \Omega} [\text{score}(\Omega, \varepsilon)] \leq E_{\varepsilon \sim \Omega'} [\text{score}(\Omega, \varepsilon)] \quad (5)$$

There are various alternative strictly proper scoring rules, such as the logarithmic scoring rule, quadratic scoring rule, and spherical scoring rule. Not only these, other strictly proper scoring rules can be generated easily due to its mathematical elegance. Meanwhile, strictly proper scoring rules are widely used for information elicitation. Truthful reporting can still be elicited even in the absence of prior information. This is similar to other concepts of incentive schemes with verification and proof, which are more powerful than classic schemes [54], [55]. Because of this, we adopt strictly proper scoring rule to design incentive schemes under the scenario without prior information (see part B in Section V).

IV. MODEL SPECIFICATION

A. SYMBOLS AND ASSUMPTIONS

In this paper, we consider a highly stylized single-period model that captures the incentive issues existing in material

planning and allocation. The timing of the events is as follows. (1) The contractors privately learn about the distribution of their respective material demands. (2) The MMD announces the allocation policy. (3) An independent mediator announces the incentive scheme to the MMD and contractors. (4) The contractors simultaneously report their material take-off to the MMD. (5) The MMD announces its inventory plan based on the reported material takeoff. (6) Material demand is realized, and allocation is made as per the announced policy. (7) The MMD and contractors are rewarded/penalized as per the announced incentive scheme. Fig. 1 illustrates the corresponding process in an incentive design setting. The notations are summarized in Table 1.

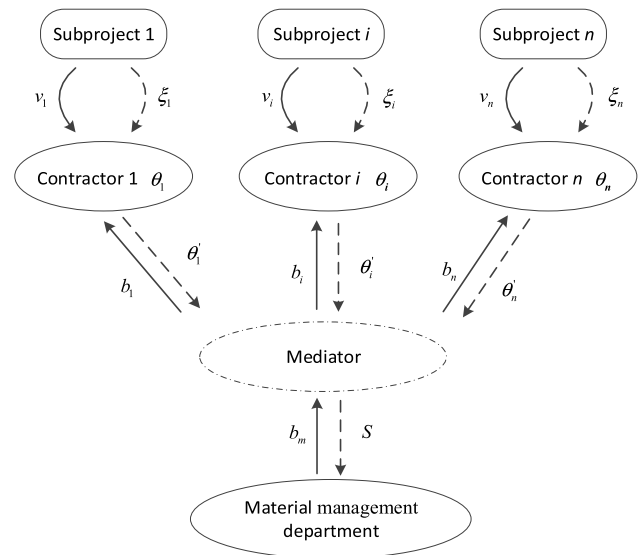


FIGURE 1. The construction supply chain in an incentive design setting.

The following assumptions are used in this paper. (1) The MMD and contractors all are risk-neutral. (2) The material demand of each contractor follows a normal distribution. (3) The mean of material demand is the private information of the corresponding contractor and other parameters are common knowledge. (4) The material-related project delay cost is undertaken by the MMD and corresponding contractor together. This is a reasonable assumption in an uncertain environment. It is difficult to determine who should be held responsible when a project delay is caused by the mismatch between material supply and demand. The best way is to bear the loss together. Meanwhile, this assumption can also stimulate the MMD and contractors to deal with the material demand and supply issues responsibly so as to reduce their individual possible loss.

B. BASIC MODEL WITHOUT INCENTIVES

In this section, a basic model is presented. This basic model denotes the problems faced by the MMD and contractors without any incentives. Based on the assumptions above,

TABLE 1. Symbols and notations.

i	Contractor or subproject $i, i=1,2,\dots,n$.
h	Material holding cost per unit material.
S	Material inventory level at the beginning of a period.
p_i	Delay cost of subproject i .
ω	The MMD's fraction of the delay cost of subprojects. Accordingly, the fraction of each contractor is $1-\omega$.
d_i	Random material demand of subproject i , with mean θ_i , variable coefficient ρ_i , probability density function $f_i(\cdot)$, and cumulative distribution function $F_i(\cdot)$.
θ'_i	Reported material takeoff of subproject i .
$l_i(S, \theta', d)$	Allocated material of subproject i .
v_i	Delay cost of subproject i .
$b_i(l, \theta', d)$	Incentive scheme of contractor i .
$b_m(l, \theta', d)$	Incentive scheme of the MMD.
$\pi_i(S, l, \theta')$	Payoff function of contractor i .
$\pi_m(S, l, \theta')$	Payoff function of the MMD.

Note that $\theta=(\theta_1, \theta_2, \dots, \theta_n)$ denotes the n -vector. $\theta_{-i}=(\theta_1, \theta_2, \dots, \theta_{i-1}, \theta_{i+1}, \dots, \theta_n)$ denotes the $n-1$ vector, except θ_i . It is the same as θ' and θ_{-i} . $d=(d_1, d_2, \dots, d_n)$ denotes the n -vector of realized demands. $(x)^+$ denotes $\max(x, 0)$. $I(\cdot)$ denotes the indicator function. If $x < y$, then $I(x < y) = 1$; otherwise, $I(x < y) = 0$.

the payoff function of contractor i can be expressed as

$$\pi_i(S, l, \theta'_i, \theta'_{-i}) = (1 - \omega) \cdot v_i \quad (6)$$

The term in Eq. (6) is a prespecified fraction of the delay cost of subproject i when the allocated material is less than the real demand. Specially, the delay cost of subproject i is given by $v_i = p_i \cdot I[l_i(S, \theta'_i, \theta'_{-i}, d) < d_i]$.

In terms of the material allocation decision, both the priority allocation policy and proportional allocation policy are taken into account. Here, the priority policy is specified as the demand-based policy. Under this policy, subprojects are assigned priorities in ascending order of their demand realizations, and then allocated material from the available inventory in decreasing order of priority. Specifically, as the material allocation decision is made after demand realizations are observed, subproject i will receive the minimum of as-yet unallocated inventory and its realized demand, for which

$$l_i(S, \theta'_i, \theta'_{-i}, d) = \begin{cases} \min\{S, d_i\}, & \phi_1 = i \\ \min\left\{S - \sum_{j=1}^{k-1} l_{\phi_j}(S, \theta'_j, \theta'_{-j}, d), d_i\right\}, & \phi_k = i \text{ \& } k \geq 2 \end{cases} \quad (7)$$

where $\{\phi_1, \phi_2, \dots, \phi_n\}$ denotes the priority list.

The proportional policy uses the information of reported material takeoff to compute the allocation fractions and thereby determine the quantity allocated to each subproject. The allocation fractions are obtained as a ratio of reported material takeoff of a subproject to total quantity of all subprojects. It should be noted that the material allocated to a particular subproject will be the minimum of the quantity

computed by this policy and its realized demand, for which

$$l_i(S, \theta'_i, \theta'_{-i}, d) = \min\left\{S \cdot \theta'_i / \sum_{j=1}^n \theta'_j, d_i\right\} \quad (8)$$

The problem faced by the contractor is to make a decision about the reported material takeoff to minimize the expected cost under the given material inventory level and allocation policy. This can be expressed as follows:

$$\min_{\theta'_i} E[\pi_i(S, l, \theta'_i, \theta'_{-i})] \quad (9)$$

Under the material allocation decision, it may mistakenly make the contractors think that the material inventory is scarce. At this point, the contractors is prone to inflate the material takeoff to gain greater allocation and protect profit. This is intuitively obvious from Eq. (9). In view of this, incentives are required for elicitation of truthful material takeoff.

On the other hand, the payoff function of the MMD is given by

$$\pi_m(S, l, \theta'_i, \theta'_{-i}) = h \cdot \left(S - \sum_{i=1}^n d_i\right)^+ + \sum_{i=1}^n \omega \cdot v_i \quad (10)$$

There are two issues in the decision of the MMD. First, the inventory planning relied heavily on the contractors' reported material takeoff. Inflated material takeoff will also misrepresent the inventory plan, and then deteriorate the system-wide performance. Second, even if the contractors report true material takeoff, the inventory level chosen by the MMD is not system-wide optimal. The construction supply chain cannot operate in optimal state. Therefore, except the incentives for contractors, an incentive is also needed to induce the MMD to adopt the system-wide optimal inventory level.

C. MODEL WITH INCENTIVES

Motivated by the above observations, we integrate decision-related cost terms into the payoff functions of the contractors and MMD. These cost schemes can be used as incentive schemes for elicitation of truthful material takeoff and system-wide optimal inventory level. For the contractors and MMD, we propose incentive schemes $b_i(l, \theta'_i, \theta'_{-i}, d)$ and $b_m(l, \theta'_i, \theta'_{-i}, d)$ respectively, which are related to the reported material takeoff θ' , realized material demands d and material allocation plan $l(S, \theta'_i, \theta'_{-i}, d)$. With the cost terms, their payoff functions can be expressed as

$$\pi_i(S, l, \theta'_i, \theta'_{-i}) = (1 - \omega) \cdot v_i + b_i(l, \theta'_i, \theta'_{-i}, d) \quad (11)$$

$$\pi_m(S, l, \theta'_i, \theta'_{-i}) = h \cdot \left(S - \sum_{i=1}^n d_i\right)^+ + \sum_{i=1}^n \omega \cdot v_i + b_m(l, \theta'_i, \theta'_{-i}, d) \quad (12)$$

Next these incentive schemes will be carefully designed such that truthful material takeoff and system-wide optimal inventory level are the optimal decisions of the contractors and MMD. These desirable requirements of incentive schemes can be characterized as truthfulness and efficiency [54]. Besides, budget balance and voluntary participation are also desirable requirements when seeking incentive schemes.

In order to ensure voluntary participation, the contractor's expected cost should not be higher than its acceptable level. This participation constraint can be expressed as

$$E[\pi_i(S, l, \theta'_i, \theta'_{-i})] \leq \pi_i^0 \quad (13)$$

where π_i^0 is the contractor's highest acceptable cost under any circumstance. Compared to the contractors, the MMD is usually more powerful and dominant in incentives design. Its expected cost can easily be ensured to be no higher than the acceptable level. Therefore, the participation constraint of the MMD is omitted.

Budget balance means that the total receipts are equal to total payments. In response, the incentive schemes of the MMD and contractors should satisfy the following relationship:

$$b_m(l, \theta'_i, \theta'_{-i}, d) + \sum_{i=1}^n b_i(l, \theta'_i, \theta'_{-i}, d) = 0 \quad (14)$$

Budget balance ensures that the system is a closed one with no surplus and no deficit [48]. Accordingly, the constructed incentive schemes can be implemented without any external supervision or external source of funding. The mediator in Figure 1 can also be omitted.

For this incentive schemes design problem, two situations are considered. In the first situation, there is prior information about the contractors' private material demand, while no prior information about the private material demand is known in the second situation. In both cases, we seek incentive schemes that satisfy the abovementioned desirable requirements

V. INCENTIVE SCHEMES

In this section, we first consider the case of having prior information about the contractors' private material demand, and develop an incentive scheme based on the Groves payment rule. Subsequently, two incentive schemes are developed on the basis of proper scoring rules in the case with no prior information.

A. INCENTIVE SCHEMES DESIGN WITH PRIOR INFORMATION

The prior information described by a distribution function represents the information that participants have about each other's private information. Assume a distribution function that represents the prior information of participants about the private material demand of contractor i , with support $\theta_i \in [\underline{\theta}_i, \bar{\theta}_i]$. In response, incentive schemes based on the Groves payment rule are developed to solve this problem.

Consider the following incentive scheme as part of the contractors' payoff function:

$$b_i(l, \theta'_i, \theta'_{-i}, d) = h \cdot \left(S - \sum_{j=1}^n d_j \right)^+ + \omega \cdot \sum_{j=1}^n v_j + (1 - \omega) \cdot \sum_{j=1, j \neq i}^n v_j - R_i(\theta) \quad (15)$$

Under this incentive scheme, the payoff of contractor i equals the system-wide total cost minus a charge scheme $R_i(\theta)$. This makes $R_i(\theta)$ a key factor for inducing contractor i

to report the true material takeoff. The following proposition provides the sufficient conditions for $R_i(\theta)$ to achieve this goal.

Proposition 1. If the following conditions 1) and 2), or conditions 1) and 3) simultaneously, satisfy a given $R_i(\theta)$, the incentive scheme defined by Eq. (15) is efficient and truthful.

- 1) $S = \underset{x}{\operatorname{argmin}} E_d \left[h \cdot (x - \sum_{i=1}^n d_i)^+ + \sum_{i=1}^n v_i \right];$
- 2) $R_i(\theta'_i, \theta_{-i}) = R_i(\theta''_i, \theta_{-i}) \quad \forall \theta'_i, \theta''_i, \theta_{-i};$
- 3) $E_{\theta_{-i}}[R_i(\theta'_i, \theta_{-i})] = E_{\theta_{-i}}[R_i(\theta''_i, \theta_{-i})] \quad \forall \theta'_i, \theta''_i.$

Proof. Under condition 1), S represents the optimal inventory level that minimizes the system-wide total cost with arbitrary expected material demand θ . If it holds, the above incentive scheme is efficient. Meanwhile, see the work of Makowski and Mezzetti [47] and Narahari et al. [48]; if conditions 1) and 2) simultaneously hold, the above incentive scheme is a Groves scheme; if conditions 1) and 3) simultaneously hold, it is a Groves scheme in expectation. In both cases, truth-telling is the dominant strategy and the Bayesian equilibrium of each contractor, respectively.

In our problem, the inventory decision can be regarded as a social choice function. If condition 1) holds, S represents the system-wide optimal inventory level, and the corresponding social choice function is considered as allocatively efficient. An allocatively efficient social choice function is one of the sufficient conditions of the Groves scheme and Groves scheme in expectation. In some studies of Groves incentive design, such as the reference [29], allocatively efficient social choice functions are inherent assumptions. But in our work, whether the inventory decision is system-wide optimal will be affected by $R_i(\theta)$. In other words, allocatively efficient social choice function isn't inherent. This makes our incentive design problem even more challenging. On the one hand, $R_i(\theta)$ should satisfy condition 1) and make the social choice function allocatively efficient. On the other hand, it should also satisfy either condition 2) or 3). Solving this issue could contribute to the application of Groves incentive.

Motivated by this, we consider a charge scheme $R_i(\theta)$ which can be expressed as

$$R_i(\theta) = Q_i - [G - G_{i+1}(\theta_{i+1}) - g(\theta) + G_i(\theta_i)] - q_i \quad (16)$$

where

$$\begin{aligned} g(\theta) &= E_d \left[h \cdot \left(S - \sum_{i=1}^n d_i \right)^+ + \sum_{i=1}^n v_i \right] \\ G &= \int_{\theta} g(\theta) d\phi(\theta) \\ G_i(\theta_i) &= \int_{\theta_{-i}} g(\theta_i, \theta_{-i}) d\phi_{-i}(\theta_{-i}) \\ Q_i &= \max_{\theta_i} \left\{ G_i(\theta_i) - E_{\theta_{-i}} E_d \left[\pi_i^0(\theta_i) \right] \right\} \\ q_i &= \beta_i \cdot \sum_{j=1}^n Q_j / n - G + T_i \\ G_{n+1}(\cdot) &= G_1(\cdot) \end{aligned}$$

In Eq. (16), $g(\theta)$ is the system-wide total cost with given θ . G is the system-wide total expected cost over all possible θ . $G_i(\theta_i)$ is the expected value of $g(\theta)$ over all possible θ_{-i} . Q_i is the minimum of $G_i(\theta_i) - \pi_i^0$ over all possible θ_i . q_i is a constant transfer payment. These parts in the charge scheme $R_i(\theta)$ are well-designed to satisfy above conditions.

Proposition 2. In the case of having prior information, no matter the priority policy and proportional policy, the incentive schemes defined by Eqs. (14), (15) and (16) is efficient and truthful.

Proof. (See Appendix).

Note that it is difficult to derive an analytical expression for the optimal inventory level due to the complexity of this inventory problem. We outline a simple and usual solution procedure to handle this case. First, the lower bound S_L and the upper bound S_U are computed via the approach proposed by Lu et al. [5]. Next, we start from $S = S_L$ and consecutively increase S by one until S_U is reached, recording the lowest expected cost of the MMD or the whole system.

Another issue with the above incentive scheme is the participation constraint of the contractors. The analysis is described in Proposition 3.

Proposition 3. If $q_i \leq 0$, then the participation constraint of contractor i is satisfied and the voluntary participation of the above incentive scheme is achieved.

Proof. (See Appendix).

In the incentive schemes defined by Eqs. (14), (15) and (16), q_i represents the constant transfer term among the MMD and contractors. Its initial role is to preserve the budget balance. In order to further preserve the voluntary participation of this incentive schemes, q_i should be nonpositive. It is equivalent that $\beta_i \cdot \sum_{j=1}^n Q_j/n + T_i \leq G$. Hence, voluntary participation can be achieved by setting appropriate β_i and T_i . Actually, such values of β_i and T_i are easy to obtain. At worst, we can make them close to zero.

Also, note that for the above incentive scheme, truth-telling is the Bayesian equilibrium strategy of each contractor. Its implementation requires that contractors share common prior information. Otherwise, it cannot work. As stated by Mallik and Harker [56], the case of limited information or even no priors is becoming increasingly relevant in real-world. At this point, designing incentive schemes without prior information is very necessary.

B. INCENTIVE SCHEMES DESIGN WITHOUT PRIOR INFORMATION

In our context, a common strictly proper scoring rule can be expressed as $\alpha \cdot (\theta' - d)^2$, where $\alpha > 0$. This can be interpreted as a misrepresentation penalty imposed for the under- and over-reporting of material demand takeoff. Specifically, the contractors are penalized for misrepresentation at the rate α . Meanwhile, if $\theta' = \theta$, then the expected penalty is the lowest. Hereinafter, we design two incentive schemes by using this strictly proper scoring rule. However, we begin by

stating the generic requirement toward the incentive schemes for inducing truth-telling of material demand takeoff.

Proposition 4. Under the situation of no prior information, contractor i will report the true material takeoff if the incentive scheme $b_i(l, \theta', d)$ satisfies the condition

$$\left\{ -(1-\omega) \cdot p_i \cdot f_i(l_i(S, \theta'_i, \theta'_{-i}, d)) \cdot \frac{\partial l_i(S, \theta'_i, \theta'_{-i}, d)}{\partial \theta'_i} + \frac{\partial b_i(l, \theta'_i, \theta'_{-i}, d)}{\partial \theta'_i} \right\} \bigg|_{\theta'_i=\theta_i} = 0 \quad (17)$$

The proof of Proposition 4 is straightforward and is omitted.

When $b_i(l, \theta', d)$ satisfies the above condition, truth-telling becomes the dominant strategy equilibrium of contractor i . Even without prior information, truth-telling can still be achieved. Also note that Eq. (17) contains the term $\partial l(S, \theta'_i, \theta'_{-i}, d)/\partial \theta'_i$. Therefore, the incentive scheme should be designed according to the definite form of material allocation policy.

Under the priority policy, contractors are ranked independently on their reported material takeoff. Theoretically, they cannot obtain more allocation by inflating the material takeoff. However, it is difficult to guarantee that contractors make decisions in a completely rational way, and irrational behavior will degrade system-wide performance. Therefore, an appropriate incentive scheme is still required to force contractors to report their true material takeoff. Based on Proposition 4, an incentive scheme (collectively called ‘the first scoring-rule-based incentive scheme’) is proposed as follows:

$$b_i(l, \theta'_i, \theta'_{-i}, d) = (1-\omega) \cdot h \cdot \left(S - \sum_{j=1}^n d_j \right)^+ \cdot \frac{l_i(S, \theta'_i, \theta'_{-i}, d)}{\sum_{j=1}^N l_j(S, \theta'_j, \theta'_{-j}, d)} + \alpha_i \cdot (\theta'_i - d_i)^2 + C_i \quad (18)$$

According to Eq.(10), we can obtain

$$b_m(l, \theta'_i, \theta'_{-i}, d) = -(1-\omega) \cdot h \cdot \left(S - \sum_{j=1}^n d_j \right)^+ - \sum_{j=1}^n \left(\alpha_i \cdot (\theta'_i - d_i)^2 + C_i \right) \quad (19)$$

The first term in Eq.(14) is a prespecified fraction, $(1-\omega) \cdot l_i(S, \theta'_i, \theta'_{-i}, d)/\sum_{j=1}^N l_j(S, \theta'_j, \theta'_{-j}, d)$, of material inventory holding cost. The second term is a misrepresentation penalty, $\alpha_i \cdot (\theta'_i - d_i)^2$, and the third term is a constant transfer payment, C_i . Under this incentive scheme, part of the inventory holding cost will be transferred from the material department to the contractors. This cost is distributed among the contractors according to their material allocation quantity.

Proposition 5. Under the priority policy, the first scoring-rule-based incentive scheme is truthful and efficient. It is also voluntary participation under a specific condition.

Proof. (See Appendix).

The specific condition mentioned in Proposition 5 refers to the values of α_i and C_i . Setting them appropriately

can achieve voluntary participation. In addition, truthfulness comes mainly because the ranking of the contractors is independent of their reported material takeoff, i.e., $\partial l_i(S, \theta'_i, \theta'_{-i}, d)/\partial \theta'_i = 0$. The contractor's decision is only affected by the misrepresentation penalty $\alpha_i \cdot (\theta'_i - d_i)^2$. He must report the true material takeoff to minimize the expected cost. On the other hand, the transferred inventory holding cost is the key factor for achieving efficiency. Through the transfer of part of the inventory holding cost, the decision problem faced by the material department is equivalent to making a decision at the inventory level to minimize the expected system-wide inventory holding cost and project delay cost.

However, under the proportional policy, the first scoring-rule-based incentive scheme fails, mainly due to the dependence of $l_i(S, \theta'_i, \theta'_{-i}, d)$ on θ'_i and θ'_{-i} . Specifically, the contractors will compete for the material allocation and their decision behaviors become very complicated. In response, another incentive scheme (collectively called 'the second scoring-rule-based incentive scheme') is proposed as follows:

$$b_i(l, \theta'_i, \theta'_{-i}, d) = -(1-\omega) \cdot v_i + \alpha_i \cdot (\theta'_i - d_i)^2 + C_i \quad (20)$$

Now, the incentive scheme for the MMD is given by

$$b_m(l, \theta'_i, \theta'_{-i}, d) = (1-\omega) \cdot \sum_{j=1}^n v_j - \sum_{j=1}^n \left(\alpha_j \cdot (\theta'_j - d_j)^2 + C_j \right) \quad (21)$$

The first term in Eq. (20) is the project delay cost transferred from the contractor to the MMD, $(1-\omega) \cdot v_i$. The second term is a misrepresentation penalty, $\alpha_i \cdot (\theta'_i - d_i)^2$, and the third term is a constant transfer payment, C_i . Note that the only difference between these two scoring-rule-based incentive schemes is the first terms in Eqs. (18) and (20). Part of the inventory holding cost is transferred from the MMD to contractors in the first scoring-rule-based incentive scheme, while the contractors' project delay costs are transferred to the material department in the second scoring-rule-based incentive scheme. In addition, Proposition 6 states the analysis about the second scoring-rule-based incentive scheme.

Proposition 6. Under the proportional policy, the second scoring-rule-based incentive scheme is truthful and efficient. It is also voluntary participation under a specific condition.

Proof. (See Appendix).

Similarly, the specific condition in Proposition 6 refers to setting the appropriate α_i and C_i . In addition to the proportional policy, this incentive scheme can also achieve the desired goals under the priority policy. The reason is straightforward. Under the second scoring-rule-based incentive scheme, the decisions of all the entities are independent of the allocation policies. Specifically, the contractors' decisions are only affected by the misrepresentation penalty, and the material department still aims to choose an inventory level to minimize the system-wide inventory holding cost and project delay cost. Hence, truthfulness, efficiency and voluntary participation can still be achieved under the priority policy.

VI. COMPARISON ANALYSIS AND NUMERICAL STUDY

A. COMPARISON ANALYSIS

Although the three incentive schemes are respectively designed under specific situations, their applicable scenarios are not restricted to them. A summary is provided in Table 2. The cases ①, ④, ⑥, ⑦, and ⑧ can be straightforwardly drawn from the analysis of previous section. In addition, the two scoring-rule-based incentive schemes can also solve the information asymmetry and coordination issue when prior information exists. Under both incentive schemes, truth-telling is the dominant strategy equilibrium of each contractor. As mentioned, a dominant strategy equilibrium can work under any prior information. Hence, we obtain cases ②, ③, and ⑤

The results in Table 2 also indicate that prior information is not necessary to achieve coordination. Whether there is prior information or not, coordination can always be achieved using the two scoring-rule-based incentive schemes. Furthermore, under the three incentive schemes, the whole system is a closed one, with no surplus and no deficit. The system-wide total cost is only affected by the inventory level and allocation policy. Under an arbitrarily material allocation policy, the system-wide total cost can always be the lowest, whether or not there is prior information and no matter which incentive scheme is implemented. This demonstrates that prior information is only the prerequisite for implementing the Groves incentive scheme and cannot improve the performance of the whole system in other aspects.

B. NUMERICAL STUDY

In this section, numerical studies are conducted to examine the performance of the proposed incentive schemes. A near-life case is investigated and the values of parameters are carefully set by following the investigation of previous

TABLE 2. Applicable scenario of each incentive scheme.

	Priority policy	Proportional policy
Prior	① Groves incentive scheme	④ Groves incentive
informat	② First scoring-rule-based	scheme
ion	incentive scheme	⑤ Second scoring-
	③ Second scoring-rule-based	rule-based incentive
	incentive scheme	scheme
No prior	⑥ First scoring-rule-based	⑧ Second scoring-
informat	incentive scheme	rule-based incentive
ion	⑦ Second scoring-rule-based	scheme
	incentive scheme	

TABLE 3. The values of the basic parameters.

h	50	p_i	[300000, 250000, 330000]
ω	0.5	ρ_i	[0.2, 0.2, 0.2]
θ_i	[120, 110, 130]	γ_i	[2,2,2]
$[\underline{\theta}_i, \bar{\theta}_i]$	[110, 130], [100, 120], [120, 140]		

TABLE 4. Decisions of each individual under priority policy and different incentive schemes.

	No incentives	Prior information			No prior information		
		Groves incentive scheme ¹	First scoring-rule-based incentive scheme ²	Second scoring-rule-based incentive scheme ³	Groves incentive scheme ¹	First scoring-rule-based incentive scheme ²	Second scoring-rule-based incentive scheme ³
Contractor 1	130	120	120	120	\	120	120
Contractor 2	120	110	110	110	\	110	110
Contractor 3	140	130	130	130	\	130	130
MMD	508	480	480	480	\	480	480

¹ $\beta_i=[0.5, 0.5, 0.5]$, $T_i=[2500, 2500, 2500]$; ² $\alpha_i=[0.5, 0.5, 0.5]$, $C_i=[-1200, -1000, -1400]$; ³ $\alpha_i=[0.5, 0.5, 0.5]$, $C_i=[-100, -100, -100]$

TABLE 5. Payoff of each individual under priority policy and different incentive schemes.

	No incentives	Prior information			No prior information		
		Groves incentive scheme ¹	First scoring-rule-based incentive scheme ²	Second scoring-rule-based incentive scheme ³	Groves incentive scheme ¹	First scoring-rule-based incentive scheme ²	Second scoring-rule-based incentive scheme ³
Contractor 1	7	160	170	190	\	170	190
Contractor 2	3	150	180	140	\	180	140
Contractor 3	12	160	220	240	\	220	240
MMD	7.413×10^3	6.126×10^3	6.026×10^3	6.026×10^3	\	6.026×10^3	6.026×10^3
Total	7.436×10^3	6.596×10^3	6.596×10^3	6.596×10^3	\	6.596×10^3	6.596×10^3

¹ $\beta_i=[0.5, 0.5, 0.5]$, $T_i=[2500, 2500, 2500]$; ² $\alpha_i=[0.5, 0.5, 0.5]$, $C_i=[-1200, -1000, -1400]$; ³ $\alpha_i=[0.5, 0.5, 0.5]$, $C_i=[-100, -100, -100]$

TABLE 6. Decisions of each individual under proportional policy and different incentive schemes.

	No incentives	Prior information			No prior information		
		Groves incentive scheme ¹	First scoring-rule-based incentive scheme	Second scoring-rule-based incentive scheme ²	Groves incentive scheme ¹	First scoring-rule-based incentive scheme	Second scoring-rule-based incentive scheme ²
Contractor 1	130	120	\	120	\	\	120
Contractor 2	120	110	\	110	\	\	110
Contractor 3	140	130	\	130	\	\	130
MMD	508	579	\	579	\	\	579

¹ $\beta_i=[0.5, 0.5, 0.5]$, $T_i=[2500, 2500, 2500]$; ² $\alpha_i=[0.5, 0.5, 0.5]$, $C_i=[-100, -100, -100]$

construction projects. Consider three contractors in this case. Their highest acceptable cost is $\pi_i^0 = \gamma_i \cdot d_i$. The prior information $\phi_i(\theta_i)$ is represented by uniform distribution. In order to reduce computational requirements, $\phi_i(\theta_i)$ is discretized with support $[\underline{\theta}_i, \bar{\theta}_i]$. The basic input parameters are presented in Table 3.

Table 4 shows the optimal decision of each individual under priority policy and different incentive schemes and Table 5 presents the corresponding cost of each individual. Observed from Tables 4, when there are no any incentives, the contractors inflate third material takeoff and the MMD

has to prepare more material inventory. Under the proposed three incentive schemes, this situation changes. The contractors report their true material takeoff, and the needed inventory decreases. Actually, 480 is the system-wide optimal inventory level. This indicates that the construction supply chain reaches the system-wide optimal state. As seen from Table 5, under those three incentive schemes, although the contractors' cost slightly increase, they are still under their highest acceptable cost. Meanwhile, the MMD's payoff also has a notable improvement compared to the case without any incentives. This is because that the inventory level is planned

TABLE 7. Payoff of each individual under proportional policy and different incentive schemes.

	No incentives	Prior information			No prior information		
		Groves incentive scheme ¹	First scoring-rule-based incentive scheme	Second scoring-rule-based incentive scheme ²	Groves incentive scheme ¹	First scoring-rule-based incentive scheme	Second scoring-rule-based incentive scheme ²
Contractor 1	27	160	\	190	\	\	190
Contractor 2	18	150	\	140	\	\	140
Contractor 3	26	160	\	240	\	\	240
MMD	1.317×10^4	1.154×10^4	\	1.144×10^4	\	\	1.144×10^4
Total	1.324×10^4	1.201×10^4	\	1.201×10^4	\	\	1.201×10^4

¹ $\beta_i = [0.5, 0.5, 0.5]$, $T_i = [2500, 2500, 2500]$; ² $\alpha_i = [0.5, 0.5, 0.5]$, $C_i = [-100, -100, -100]$

based on the true material takeoff. More than that, performance of the total system decreases nearly 11.3%. In the light of this, a win-win coordination is reached by the proposed three incentive schemes.

Table 6 shows the optimal decision of each individual under proportional policy and different incentive schemes and Table 7 presents the corresponding cost of each individual. The results show that the Groves incentive scheme and second scoring-rule-based incentive scheme can solve the information asymmetry and coordination issues under proportional policy. They can enable the win-win coordination along the construction supply chain. Meanwhile, the first scoring-rule-based incentive scheme fails to do this under proportional policy. All the similar results seen from Tables 4-7 confirm the applicable scenario of each incentive scheme, as presented by Table 2.

C. MANAGERIAL IMPLICATIONS

According to the comparison analysis and numerical study, several major managerial implications are obtained.

First, implementing incentives can protect MMD's payoff, either the Groves incentive scheme or the two scoring-rule-based incentive schemes. When no incentives, the contractors may misrepresent the material demand for reducing costs close to 0. Under this situation, MMD bears all the cost at a very high level. But when an incentive scheme is implemented, no matter which one, this situation changes. As shown in the numerical studies, MMD's cost has been significantly reduced. Meanwhile, the contractors' costs still remain at a negligible level. This guarantees their willingness to participate in the incentive schemes.

Second, from the point of applicability, the second scoring-rule-based incentive scheme is a better choice than the other incentive schemes. This view can be explained from two aspects. On the one hand, intuitively, the mathematical models of the two scoring-rule-based incentive schemes are simpler than the Groves incentive scheme. This makes them easier to be understood and implemented in practice. On the other hand, as mentioned, the prior information is only a prerequisite for implementing the Groves incentive scheme and

has no other special meaning. When there is no prior information, the two scoring-rule-based incentive schemes can be adapted to address the information asymmetry and coordination issues. Especially, the second scoring-rule-based incentive scheme can also achieve coordination no matter which allocation policy is adopted. As shown in Table 2, it can be used to address the information asymmetry and coordination issues under all the four scenarios. At this point, by using the second scoring-rule-based incentive scheme, MMD does not need to pay extra time and cost to obtain the prior information on material demands, nor worry too much about the allocation policies. It will be very easy to implement the incentive.

VII. CONCLUSION

In this paper, we investigate the incentive schemes for centralized material planning and allocation with asymmetric information. Three incentive schemes are designed under different scenarios regarding prior information, as well as with the goals of truthfulness, efficiency, and voluntary participation. Under the case of having prior information, a Groves incentive scheme size designed, while under the case of no prior information, two scoring-rule-based incentive schemes are designed. By examining the optimal decisions of MMD and contractors under the three incentive schemes, it is found that all of them can achieve the desired goals of truthfulness, efficiency, and voluntary participation. Meanwhile, the comparison analysis and numerical study demonstrate that the prior information is only a prerequisite for implementing the Groves incentive scheme and cannot improve the system performance. The results also show that the second scoring-rule-based incentive scheme is more recommendable from the point of implementation simplicity. It is not only simple in mathematical form, but also can achieve coordination without prior information and under different allocation policy.

This research raises several relevant problems for further consideration. First, risk attitudes are not considered in this paper. In practice, the MMD and contractors may be risk-averse or risk-taking. The way they make decisions will change accordingly. Thus, future studies need to investigate

incentive schemes with a consideration of risk attitudes. Specifically, the popular Mean-Variance (MA), Value-at-Risk (VaR), and Conditional VaR (CVaR) can be used to model their payoff under different risk attitudes [57]. Second, there may be other incentive schemes between the MMD and contractors, such as the material waste reduction incentive scheme. The incentive schemes in this paper and the waste reduction incentive scheme may intertwine and produce some complicated effects on the material planning and operations. Therefore, it would be valuable to extend the problem using this incentive scheme consideration and investigate their interaction.

APPENDIX

A. PROOF OF PROPOSITION 2

Under the incentive scheme defined by Eqs. (14), (15) and (16), the expected cost of the MMD is given by

$$\begin{aligned} E_d [\pi_m (S, l, \theta'_i, \theta'_{-i})] \\ = E_d \left[h \cdot \left(S - \sum_{i=1}^n d_i \right)^+ + \sum_{i=1}^n v_i \right] \\ + \left(1 - \sum_{i=1}^n \beta_i / n \right) \sum_{i=1}^n Q_i - \sum_{i=1}^n T_i \end{aligned}$$

Note that Q_i and T_i are independent of S . Under both allocation policies, the problem faced by the MMD is equivalent to making a decision at the inventory level to minimize the expected system-wide total cost. Therefore, the incentive scheme is efficient and condition 1) in Proposition 1 holds.

Furthermore, the four terms in the brackets of Eq. (16) cancel each other with the expectation of θ_{-i} , which is given by

$$E_{\theta_{-i}} [R_i (\theta, d)] = Q_i - [G - G - G_i (\theta_i) + G_i (\theta_i)] - q_i$$

As Q_i and q_i are independent from θ_i and l_i , the incentive scheme satisfies condition 3) of Proposition 1 under both allocation policies. As can be seen from Proposition 1, this incentive scheme is a Groves scheme in expectation. Truth-telling is the Bayesian Nash equilibrium of each contractor.

B. PROOF OF PROPOSITION 3

Under the Groves incentive scheme in expectation defined by Eqs. (14), (15) and (16), the expected cost of contractor i is given by

$$\begin{aligned} E_d [\pi_i (S, l, \theta'_i, \theta'_{-i})] = g(\theta) - Q_i + [G - G_{i+1}(\theta_{i+1}) \\ - g(\theta) + G_i(\theta_i)] + q_i \end{aligned}$$

The expected cost with respect to θ_{-i} is given by

$$E_{\theta_{-i}} E_d [\pi_i (S, \theta)] = G_i (\theta_i) - Q_i + q_i$$

As $Q_i \geq G_i(\theta_i) - \pi_i^0$, we obtain

$$E_{\theta_{-i}} E_d [\pi_i (S, \theta)] \leq \pi_i^0 + q_i$$

According to this equation, we find that the participation constraint of contractor i can be satisfied when $q_i \leq 0$.

C. PROOF OF PROPOSITION 5

Under the priority policy, we obtain $\partial l_i(S, \theta'_i, \theta'_{-i}, d) / \partial \theta'_i = 0$. At this point, with regard to the first scoring-rule-based incentive scheme, we obtain $\partial b_i(l, \theta'_i, \theta'_{-i}, d) / \partial \theta'_i = 2\alpha_i \cdot (\theta'_i - \mu_i)$. The condition shown as Eq. (17) is satisfied. Therefore, contractor i will report the true material takeoff.

Under this incentive scheme, the expected cost of the MMD can be expressed as

$$\begin{aligned} E_d [\pi_m (S, l, \theta'_i, \theta'_{-i})] \\ = E_d \left\{ \omega \cdot \left[h \cdot \left(S - \sum_{i=1}^n d_i \right)^+ + \sum_{i=1}^n v_i \right] \right\} \\ - E_d \left\{ \sum_{i=1}^n \left[\alpha_i \cdot (\theta'_i - d_i)^2 + C_i \right] \right\} \end{aligned}$$

Note that $\alpha_i \cdot (\theta'_i - d_i)^2 + C_i$ is independent of S . The problem faced by the MMD is equivalent to making a decision at the inventory level to minimize the expected system-wide total cost. Therefore, this incentive scheme is efficient.

Lastly, the expected cost of contractor i under the first scoring-rule-based incentive scheme is given by

$$E_d [\pi_i (S, l, \theta'_i, \theta'_{-i})] = Z_0 \cdot \alpha_i + C_i + Z_1$$

where

$$\begin{aligned} Z_0 &= E_{d_i} \left[(\theta_i - d_i)^2 \right] \\ Z_1 &= E_d \left[(1 - \omega) \cdot \left(v_i + h \cdot \left(S - \sum_{j=1}^n d_j \right)^+ \right. \right. \\ &\quad \left. \left. \cdot \frac{l_i (S, \theta'_i, \theta'_{-i}, d)}{\sum_{j=1}^n l_j (S, \theta'_j, \theta'_{-j}, d)} \right) \right] \end{aligned}$$

If $Z_0 \cdot \alpha_i + C_i + Z_1 \leq \pi_i^0$, then the participation constraint of contractor i is satisfied. Intuitively, there is a large number of α_i and C_i that satisfies this condition. Hence, we can easily obtain a voluntary participation incentive scheme.

D. PROOF OF PROPOSITION 6

Under the proportional policy and second scoring-rule-based incentive scheme, we obtain $\partial E[\pi_i(S, l, \theta'_i, \theta'_{-i})] / \partial \theta'_i = 2\alpha_i \cdot (\theta'_i - \mu_i)$. Reporting true material takeoff is the optimal choice for contractor i . Therefore, this incentive scheme is truthful.

In addition, the expected cost to the MMD under this incentive scheme is given by

$$\begin{aligned} E_d [\pi_m (S, l, \theta'_i, \theta'_{-i})] = E_d \left[h \cdot \left(S - \sum_{i=1}^n d_i \right)^+ + \sum_{i=1}^n v_i \right] \\ - E_d \sum_{i=1}^n \left[\alpha_i \cdot (\theta'_i - d_i)^2 + C_i \right] \end{aligned}$$

As $\alpha_i \cdot (\theta'_i - d_i)^2 + C_i$ is independent of S , the problem faced by the MMD is equivalent to making a decision at the inventory level to minimize the expected system-wide total cost. Hence, this incentive scheme is efficient.

Lastly, the expected cost to contractor i can be expressed as

$$E_d [\pi_i (S, l, \theta'_i, \theta'_{-i})] = Z_0 \cdot \alpha_i + C_i$$

where

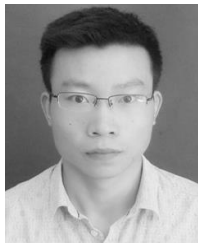
$$Z_0 = E_{d_i} \left[(\theta'_i - d_i)^2 \right]$$

If $Z_0 \cdot \alpha_i + C_i \leq \pi_i^0$, then this incentive scheme is participatory. Similar to the previous analysis, there is also a large number of α_i and C_i that satisfy this condition.

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